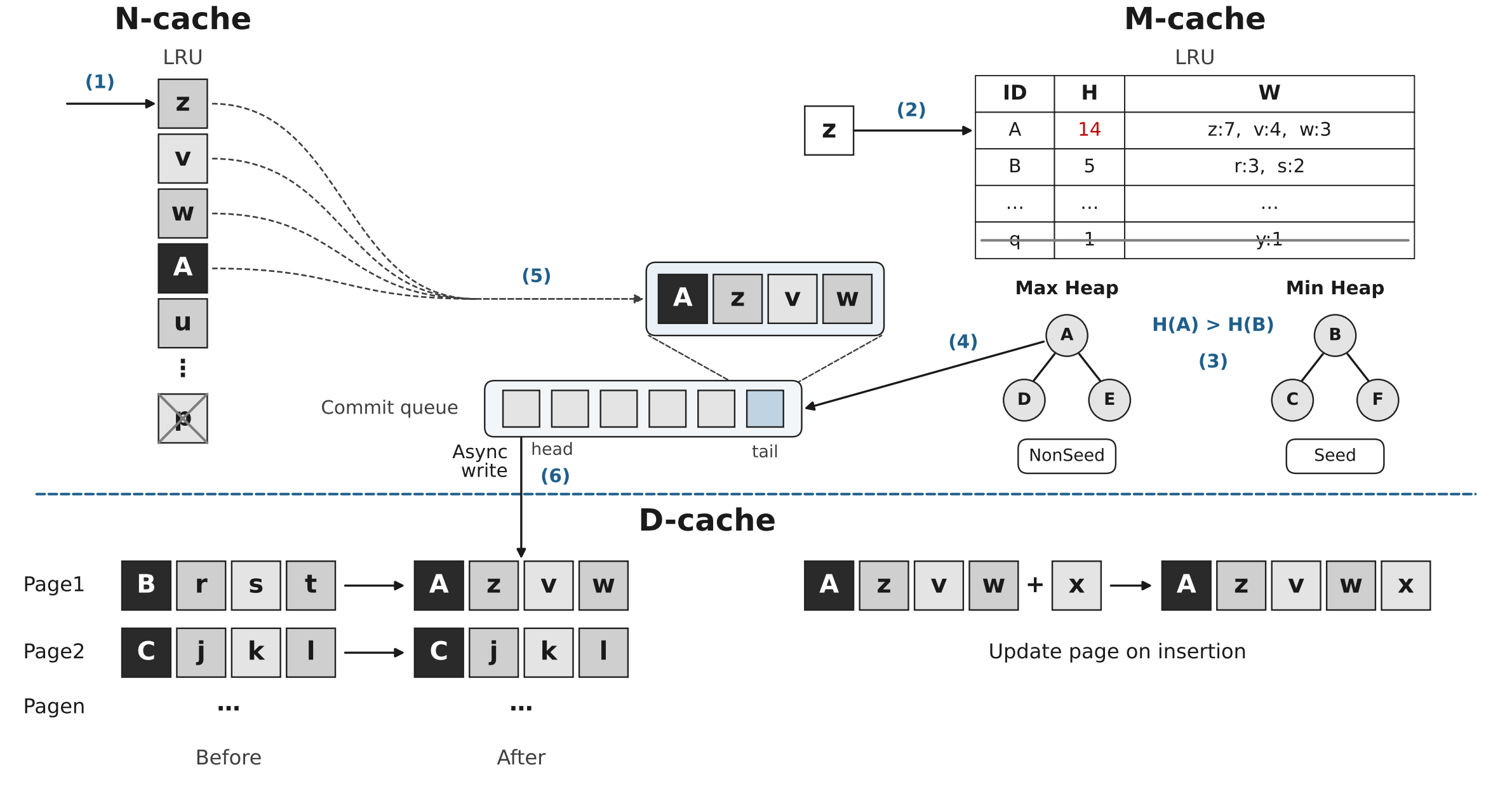
N-cache
LRU
(1)
z
v
w
A
u
p
M-cache
LRU
z
(2)
ID
H
W
A
14
z:7, v:4, w:3
B
5
r:3, s:2
...
...
...
q
1
y:1
(5)
A
z
v
w
Max Heap
Min Heap
H(A) > H(B)
(3)
(4)
A
D
E
B
C
F
NonSeed
Seed
Commit queue
Async write
head
tail
(6)
D-cache
Page1
B
r
s
t
A
z
v
w
Page2
C
j
k
l
C
j
k
l
Pagen
...
...
Before
After
A
z
v
w
+
x
A
z
v
w
x
Update page on insertion